

Unit 0 Project Brief

Aerospace Readiness Mini-Challenge

Mission Readiness: Safety, Tools & Engineering Systems

Challenge Statement: Design, build, test, and document a small aerospace support system that solves a simple classroom mission need while following all safety, certification, documentation, and teamwork expectations.

Project Overview

In this launch project, your team demonstrates that you are ready to work safely, document engineering decisions, use shared FabLab tools responsibly, and think about aerospace systems as connected subsystems. The build is intentionally small; the focus is on readiness, process, and evidence.

Mission Scenario

Aerospace teams do not begin major missions by building the largest system first. They verify procedures, roles, tools, communication, and readiness through smaller tasks before moving into more complex work. Your team will complete a mini-challenge that proves you are prepared for future POE projects.

Design Requirements

- Use an approved aerospace support problem such as a drone landing marker, payload fixture, rocket component stand, VEX test base, or motion-test fixture.
- Complete or confirm required shared FabLab certifications before using controlled tools independently.
- Create a clear problem statement, criteria, constraints, and at least one labeled concept sketch.
- Use safe tool habits, correct PPE, and clean workspace procedures throughout the activity.
- Build a small physical prototype or fixture using approved classroom materials.
- Test whether the system meets its intended purpose and document at least one improvement recommendation.

Constraints

- Project must stay small enough to complete during the Unit 0 readiness window.
- Only approved materials and tools may be used.
- No independent laser, 3D printer, or hand tool use until the correct certification is complete.
- Prototype must not create sharp edges, unstable supports, unsafe moving parts, or unsafe drone/rocket conditions.
- Final evidence must be documented in the engineering notebook or assigned digital format.

Required Engineering Evidence

Problem Definition	Problem statement, criteria, constraints, and selected mini-challenge.
Design Evidence	Labeled sketch, planned materials, and explanation of how the system is supposed to work.
Safety Evidence	Certification status, PPE/tool notes, and safe work habits.
Testing Evidence	Photo, observation notes, simple measurement, pass/fail check, or user feedback.

Reflection	What worked, what failed, and what you would improve before a larger aerospace project.
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Project Checkpoints

- 1 Certifications confirmed
- 2 Problem and sketch approved
- 3 Prototype built safely
- 4 Test completed
- 5 Final readiness reflection submitted

Success Criteria

Readiness & Safety	Certifications, PPE, tool habits, and workspace expectations are followed consistently.
Design Thinking	Problem, criteria, constraints, and concept are clear and connected to an aerospace need.
Prototype & Test	Prototype is built safely and tested against its intended purpose.
Documentation	Notebook evidence is complete, organized, and useful for future engineering work.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.